CLOMIFENE

Multiple pregnancy risk

Ovarian hyperstimulation

CHEMICAL NAME (E/Z)-2-[4-(2-chloro-1,2-diphenylethenyl)phenoxy]-N,N-diethylethanamine (citrate)

CLASS Selective oestrogen receptor modulator (ovulation inducer)

TRADE NAMES Clomid · Serophene · Clomiphene citrate

THERAPEUTIC USES / INDICATIONS

- Anovulatory infertility, especially polycystic ovary syndrome (letrozole increasingly preferred).
- Male hypogonadism (off-label).

ADMINISTRATION

- **PO** daily for 5 days starting day 2–5 of the cycle.
- Ovulation monitoring (ultrasound or day-21 progesterone) recommended.
- Maximum 6 cycles (ovarian cancer concern).

DOSAGES

- **Start:** 50 mg daily for 5 days.
- If no ovulation, increase to 100 mg daily for 5 days next cycle (do not exceed 100 mg/day for 5 days).
- Stop after about 6 cycles in total (re-evaluate if no ovulation after 3 courses).

PHARMACOKINETICS

- **Onset:** ovulation 5–10 days after last dose.
- Metabolised in liver; enterohepatic recirculation.
- Excreted in faeces. **t½:** 5–7 days.

PHARMACODYNAMICS

- Blocks **hypothalamic oestrogen receptors** → removes negative feedback → ↑ GnRH, FSH, LH → follicle development and ovulation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hot flushes.
- Abdominal discomfort, bloating, ovarian enlargement.
- Breast tenderness, nausea, mood change.

ADVERSE EFFECTS (SERIOUS)

- Ovarian hyperstimulation syndrome (OHSS).
- Multiple pregnancy (~5–10% twins).
- Visual disturbances (blurring, spots, flashes); stop permanently.
- Ectopic pregnancy; thinning of endometrium.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy.
- Liver disease; abnormal uterine bleeding of unknown cause.
- Ovarian cysts (non-PCOS); hormone-dependent tumours.
- Uncontrolled thyroid/adrenal disease.

KEY INTERACTIONS

- No significant drug interactions.

NURSING MANAGEMENT

- Explain timing of tablets and intercourse (days 10–17).
- Arrange ultrasound/progesterone monitoring.
- Ask about visual symptoms and abdominal pain.
- Exclude pregnancy before each cycle.

PATIENT EDUCATION

- Take for 5 days at the start of your cycle as instructed.
- Stop and report blurred vision or flashing lights.
- Report severe abdominal pain, bloating, breathlessness (OHSS).
- Risk of twins.

KEY POINTS

- SERM — induces ovulation.
- Multiple pregnancy, OHSS, visual disturbance.
- 50 mg × 5 days from day 2–5.
- Max 6 cycles.

SOURCES CHECKED **[690]** Clomifene 50mg Tablets (UK SmPC) **[691]** Clomiphene citrate, oral (US label) **[692]** Clomiphene Citrate, oral (US label) *Full references p. 432–457*